

Chapter 3: Diversity in Living Things and their Classification

Textbook Exercise Solutions & Answers

1. Match the pairs.

Column 'A' Column 'B' 1. Amphibians a. A monkey 2. Vertebrates b. A snake 3. With scales c. A frog

Column 'A'	Column 'B'
1. Amphibians	c. A frog
2. Vertebrates	a. A monkey
3. With scales	b. A snake

2. Who is the odd one out?

(a) Fungus, mushroom, chrysanthemum, spirogyra

Chrysanthemum (only flowering plants)

or

Spirogyra (only aquatic plant)

(b) Mango, banyan, palm, chickpea

Chickpea (shrub, others are not)

or

Palm (tall tree without any branches, others have branches)

(c) Grape, orange, lemon, hibiscus

Grape (only creeper, others are not)

or

Hibiscus (flower, others are fruits)

(d) Sunflower, banyan, jowar, bajra

Banyan (perennial, others are not)

(e) Guava, radish, carrot, beetroot

Guava (fruit, others are roots)

(f) Deer, fish, man, worms

Worms (invertebrate, others are not)

or

Fish (aquatic animal, others are terrestrial animals)

3. What is the difference?

(a) Flowering plant and Non-flowering plant

Flowering plant	Non-flowering plant
1. Plants that bear flowers are called flowering plants.	1. Plants that do not bear flowers are called nonflowering plants.
2. They have roots, stems and leaves.	2. They may not have roots, stems or leaves.
e.g. Rose, sunflower, lotus	e.g. Pine, fern, algae

(b) A Tree and a Shrub

Tree	Shrub
1. They have branches at some height above the ground.	1. They give out branches close to the ground.
2. They grow very tall compared to shrubs.	2. They grow upto 2 to 3 metres in height.
3. They have strong and hard stems.	3. They have thick hard stems.
e.g. Mango tree, banyan tree	e.g. Hibiscus, oleander

(c) Vertebrates and Invertebrates

Vertebrates	Invertebrates
1. They have vertebral column.	1. They do not have vertebral column.
2. The brain is well developed.	2. The brain if present may not be developed.
e.g. Human, dog, fishes etc.	e.g. earthworm, insects etc.

4. True or false?

(a) The snail is an aquatic animal.

False

(b) Amphibians can live in air and in water.

False

(c) The function of the brain is well developed in vertebrate.

True

(d) The amoeba is a multicellular animal.

False

5. Write two names of each.

(a) A flowering plant

Rose, hibiscus

(b) A non-flowering plant

Fern, algae

(c) A tree

Mango, banyan

(d) A shrub

Oleander, hibiscus

(e) A creeper

Pumpkin, watermelon

(f) An annual plant

Jowar, sunflower

(g) A biennial plant

Carrot, beetroot

(h) A perennial plant

Mango, gulmohar

6. Write answers to the following.

(a) What are the parts of a plant?

The parts of a plant are: root, stem, leaves, flowers, fruit and seed.

(b) What are the functions of root?

The functions of root are:

1. The root hold the soil firmly and anchor the plant.
2. They absorb water, salt and minerals from the soil and transport it to stem.
3. Some roots store food. e.g. Carrot, radish.

(c) Why is it necessary to classify living things?

1. There is a great diversity in living things- both plants and animals on the earth.
2. In order to identify them and to study their characteristics, systematic classification is necessary.

(d) What are the criteria used to classify living things?

Following are the criteria used to classify living things:

a. Plants:

- Mode of nutrition.
- Similarities and differences in structure.
- Presence of flowers or not.
- Period of life cycles.
- Habitat.

b. Animals:

- Cell structure.
- Presence of vertebral column.
- Method of reproduction.
- Habitat.

(e) Tell some characteristics of creepers.

The characteristics of creepers are:

1. The stem of a creeper is very flexible, soft and green and need support to climb.
2. Some climbers spread on the ground, e.g. Pumpkin
3. Some have aerial roots, e.g. Money plant
4. Some climbers develop tendrils, e.g. Cucumber

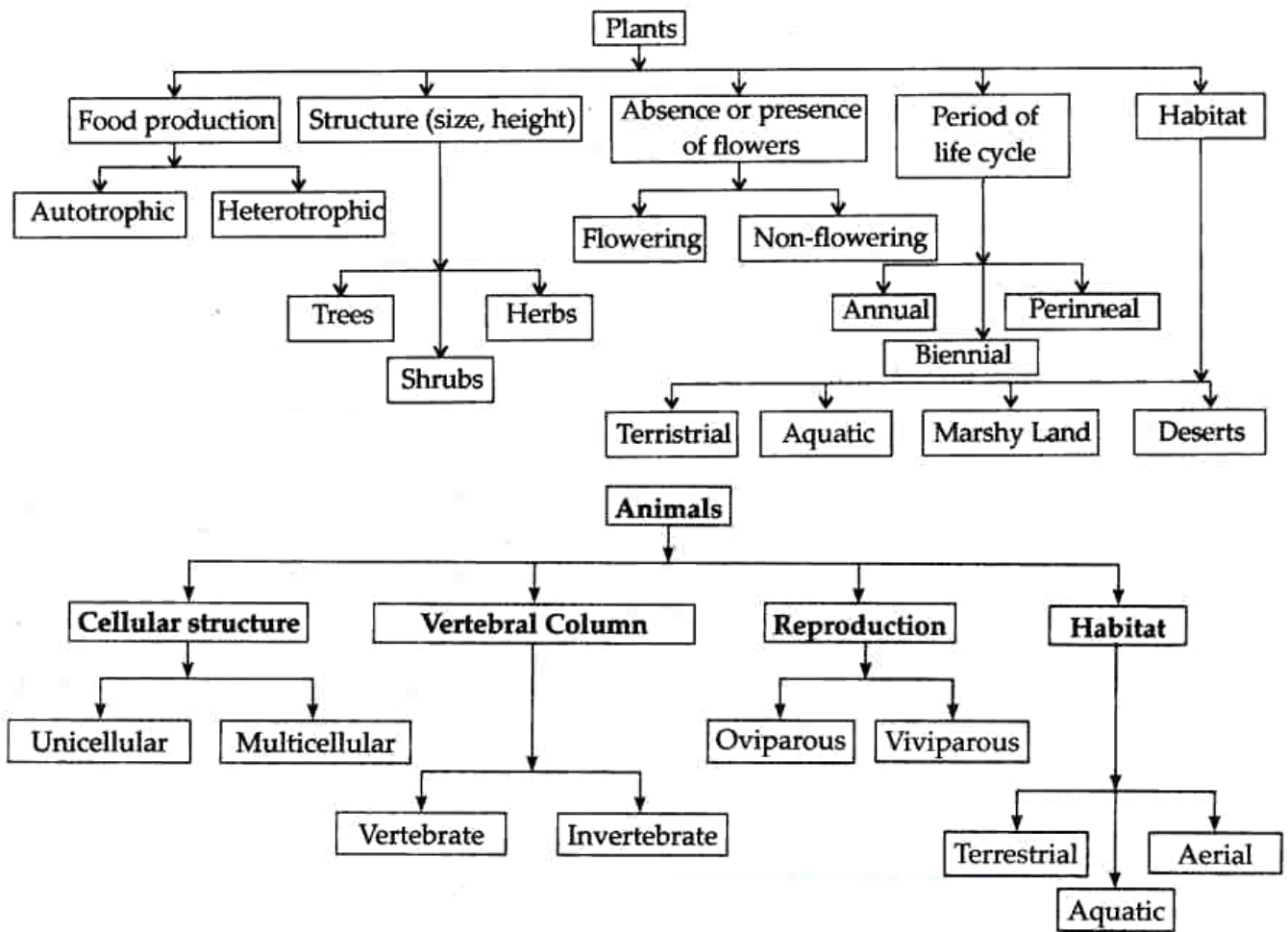
(f) Explain the characteristics of herbs with examples.

1. Herbs grow 1 to 1.5 m tall.
2. Their stem is green and quite flexible.
3. They may live for a few months or upto two years, e.g. Tulsi, oregano, etc.

(g) How is the body of animals protected?

Body of animals are protected by special covering such as skins, feathers, scales, hair fur, etc.

(h) On the basis of which criteria will you classify plants and animals?



7. Draw figures.

Draw the figure of a plant to show the parts, namely, the root, stem and leaves in it.

